

Sai Saree Academy — Backend API Specification

Comprehensive Architecture & Endpoint Briefing Report

1. System Design & Database Schema

The backend is developed with Next.js 16 App Router using dynamic route handlers. Data storage is powered by PostgreSQL hosted on Neon DB and modeled through Prisma ORM.

- Relational Model: Defines Users, Events, and Registrations. Users and Events possess a many-to-many relationship map resolved through the Registrations junction table. Cascade delete triggers ensure that removing events or users automatically cleans up associated registrations.

2. Key Automation Pipelines

- Google Meet Integration: Updates payment_status to COMPLETED and emails class details containing the Google Meet joining URL.
- PDF Certificate Generation: Automatically compiles landscape A4 PDFs overlaying user name and workshop date. Employs a fallback pattern to run procedurally if custom file 'certificate-template.pdf' is missing.
- Lazy Permission Revocation: Drive permissions are granted for 30 days. To eliminate expensive cron processes, expiration checks are lazy-evaluated on direct recordings endpoints and registration lists, cleaning up Google Drive folder permissions on-the-fly.

3. Public & Authentication Endpoints

[POST] /api/auth/signup Access: Public
Desc: Registers new user accounts. Inputs are sanitized and password credentials are hashed via Bcryptjs. Role parameter defaults to 'user'. Request: JSON { name, email, phone, password, role } Success Response: JSON { message, user: { id, name, email } }
[POST] /api/auth/login Access: Public

Desc: Authenticates users and signs a secure JWT payload containing user metadata. JWT is written directly to the client's cookie storage.

Request: JSON { email, password }

Success Response: Set-Cookie: token=<jwt_token> | JSON { message, user }

4. User-Facing Booking & Access Endpoints

Allows registered clients to discover classes, manage profiles, enroll in workshops, and access certificates and recordings.

[GET] /api/events

Access: Public

Desc: Lists all courses and saree pleating workshops created by admins, sorted chronologically.

Request: None

Success Response: JSON Array [{ id, title, description, date, time, location, meet_link, drive_folder_id }]

[POST] /api/events/register

Access: Logged-in User

Desc: Enrolls the logged-in user in a specific workshop. Sets payment status to PENDING.

Request: JSON { event_id }

Success Response: JSON { message, registration }

[GET] /api/events/my-registrations

Access: Logged-in User

Desc: Displays a summary dashboard of all classes bookmarked or registered by the user. Automatically calls the lazy Google Drive access check to ensure folder links are revoked if expired.

Request: None

Success Response: JSON Array [{ event_id, event_title, payment_status, attendance_status, meet_link, drive_folder_link }]

[GET] /api/events/recordings

Access: Logged-in User

Desc: Direct link query to fetch recordings. Evaluates if the 30-day Google Drive folder share period is still valid. Triggers lazy revocation.

Request: Query Param: ?event_id=<id>

Success Response: JSON { drive_folder_link: 'url' }

[PUT] /api/users/update

Access: Logged-in User

Desc: Allows users to modify profile information. Validates phone numbers and prevents taking emails/phones belonging to other active profiles.

Request: JSON { name, email, phone }

Success Response: JSON { message, user }

5. Admin Event Administration & CRUD

Administrative actions require authorization. Routes inspect user credentials decoded from the JWT token cookie and block requests if the user lacks the 'admin' role.

[POST] /api/admin/events/create

Access: Admin Only

Desc: Creates a new class event, establishing dates, location boundaries, and pricing metadata.

Request: JSON { title, description, date, time, location, images, registration_deadline }

Success Response: JSON { message, event }

[PUT] /api/admin/events/update

Access: Admin Only

Desc: Updates event properties, including assigning Google Meet join links and linking Google Drive recording folders.

Request: JSON { id, title, location, date, meet_link, drive_folder_id }

Success Response: JSON { message, updatedEvent }

[DELETE] /api/admin/events/delete

Access: Admin Only

Desc: Removes an event and its related registration bookings.

Request: JSON { id }

Success Response: JSON { message, deletedEvent }

[GET] /api/admin/events/view

Access: Admin Only

Desc: Fetches detailed operational statistics for a single workshop, enumerating enrollment, paid ratios, attendance rates, and active permissions.

Request: Query Param: ?id=<id>

Success Response: JSON { event, stats: { total_registrations, paid_count, present_count }, registrations: [...] }

[GET] /api/admin/events/viewregistrations

Access: Admin Only

Desc: Returns a master list of all bookings across all workshops in the system. Runs lazy Drive access evaluation.

Request: None

Success Response: JSON Array [{ user_id, event_id, payment_status, attendance_status, user: {...}, event: {...} }]

6. Admin Automation & Post-Event Pipelines

Automates payment confirmations, joining links, credential dispatch, and recorded material access management.

[POST] /api/admin/events/approve-payment

Access: Admin Only

Desc: Confirms user payment, updating status to COMPLETED, and automatically triggers an email notification delivering the Google Meet join URL.

Request: JSON { event_id, user_ids: [id1, id2] }

Success Response: JSON { message, updated_count }

[POST] /api/admin/events/mark-attendance

Access: Admin Only

Desc: Allows bulk logging of participant attendance status (PRESENT, ABSENT, or NOT_MARKED).

Request: JSON { event_id, attendance: [{ user_id, status }] }

Success Response: JSON { message, results: [{ user_id, status, success }] }

[POST] /api/admin/events/send-certificates

Access: Admin Only

Desc: Processes certificate dispatches. Finds all participants who have COMPLETED payments and are marked PRESENT, builds custom landscape PDFs, emails them, and sets 'certificate_sent' to true.

Request: JSON { event_id }

Success Response: JSON { message, processed_count, skipped_count }

[POST] /api/admin/events/grant-drive-access

Access: Admin Only

Desc: Shares the event's recordings folder with eligible attendees (paid + present) via Google Drive API (Writer/Reader), storing a 30-day expiry timestamp.

Request: JSON { event_id }

Success Response: JSON { message, granted_count }

[DELETE] /api/admin/users/delete

Access: Admin Only

Desc: Permanently deletes user accounts and registration data. Guarded to prevent admins from deleting their own logged-in admin profiles.

Request: JSON { id }

Success Response: JSON { message, deletedUser }